

P2 — Sustainable Energy

OCR GCSE Physics B (J259) — 21st Century Science · Paper 1 (Breadth in Physics)

P2.1 Energy Stores and Transfers

Energy cannot be created or destroyed — it is only **transferred** between stores (the **Principle of Conservation of Energy**). What we call "using energy" really means *transferring* energy from a useful store to less useful ones (often into the thermal store of the surroundings).

The Eight Energy Stores

Store	Example
Kinetic	Anything moving — a running athlete, a flying cricket ball
Gravitational potential	An object raised in a gravitational field — a book on a shelf
Elastic potential	A stretched spring, a drawn bow, a rubber band
Thermal (internal)	A hot cup of tea, the Sun
Chemical	Food, fuels, batteries
Magnetic	Two repelling magnets held apart
Electrostatic	Charged Van de Graaff, thunderclouds
Nuclear	Stored in the nuclei of atoms (fission/fusion fuels)

The Four Pathways

Energy is moved *between* stores by four pathways:

- **Mechanical work** — a force doing work (pushing, lifting, stretching).
- **Electrical work** — charges being moved by a p.d.
- **Heating** — energy transferred from hot to cold (particle collisions, radiation).
- **Radiation** — energy carried by waves (light, sound, IR).

P2.2 Calculating Energy Changes

Kinetic energy: $E_k = \frac{1}{2} m v^2$

$$(J) = (kg) \times (m/s)^2$$

Gravitational potential energy: $E_p = m g h$

$$(J) = (kg) \times (N/kg) \times (m) \quad (g \approx 9.8 \text{ N/kg on Earth})$$

Elastic potential energy: $E_e = \frac{1}{2} k x^2$

$$(J) = (N/m) \times (m)^2$$

Heating (change in thermal store): $\Delta E = m c \Delta\theta$

$$(J) = (kg) \times (J/kg \text{ } ^\circ\text{C}) \times (^\circ\text{C})$$

Worked Example: A 70 kg cyclist is moving at 8 m/s. How much kinetic energy do they have?

$$E_k = \frac{1}{2} \times 70 \times 8^2 = \frac{1}{2} \times 70 \times 64 = \mathbf{2240 \text{ J}}$$

Worked Example: A 2 kg book is lifted 1.5 m onto a shelf. How much GPE does it gain?

$$E_p = m g h = 2 \times 9.8 \times 1.5 = \mathbf{29.4 \text{ J}}$$

P2.3 Efficiency and Sankey Diagrams

The **efficiency** of a device is the fraction (or %) of the input energy that is transferred **usefully**. The rest is *wasted* — usually to the thermal store of the surroundings.

$$\begin{aligned}\text{Efficiency} &= \text{useful energy output} / \text{total energy input} \\ &= \text{useful power output} / \text{total power input}\end{aligned}$$

Sankey Diagram — Filament Lamp (100 J input)



Efficiency = $10/100 = 0.10 = 10\%$ — that's why LEDs ($\approx 30\text{--}50\%$) have replaced filament bulbs.

Worked Example: A motor lifts a 5 kg load 2 m. If the motor uses 200 J of electricity, what is its efficiency?

Useful output = $m g h = 5 \times 9.8 \times 2 = 98 \text{ J}$. Efficiency = $98/200 = \mathbf{0.49 (49\%)}$

P2.4 Power

Power is the rate at which energy is transferred (or the rate at which work is done). Unit: **watts (W)**, where $1 \text{ W} = 1 \text{ J/s}$.

$$P = E / t \quad \text{or} \quad P = W / t$$

$$(W) = (J) / (s)$$

Worked Example: A hairdryer transfers 36 000 J in 30 s. What is its power?

$$P = E / t = 36\,000 / 30 = \mathbf{1200\,W\ (1.2\,kW)}$$

P2.5 Reducing Unwanted Energy Transfers

In our homes, most wasted energy leaves through the walls, roof, windows and floors as heat. We can reduce this with:

Method	How it works
Loft insulation (fibreglass)	Traps air — a poor conductor — reducing conduction through the ceiling
Cavity wall insulation (foam)	Reduces conduction and convection currents inside the wall cavity
Double glazing	The gas/vacuum gap between the panes is a very poor conductor
Draught excluders	Prevent heat loss by air exchange (convection)
Thicker walls / lower thermal conductivity	Reduce the rate of heat conduction

Rate of cooling depends on: thickness, thermal conductivity, surface area, and temperature difference between inside and outside.

Exam Tip: Questions often ask about *payback time*: time = cost of insulation / annual saving. A £200 loft insulation that saves £50 a year pays back in 4 years.

P2.6 Energy Resources

An **energy resource** is a naturally occurring store from which useful energy can be extracted — usually to make electricity, heat buildings or move vehicles.

Renewable vs Non-Renewable

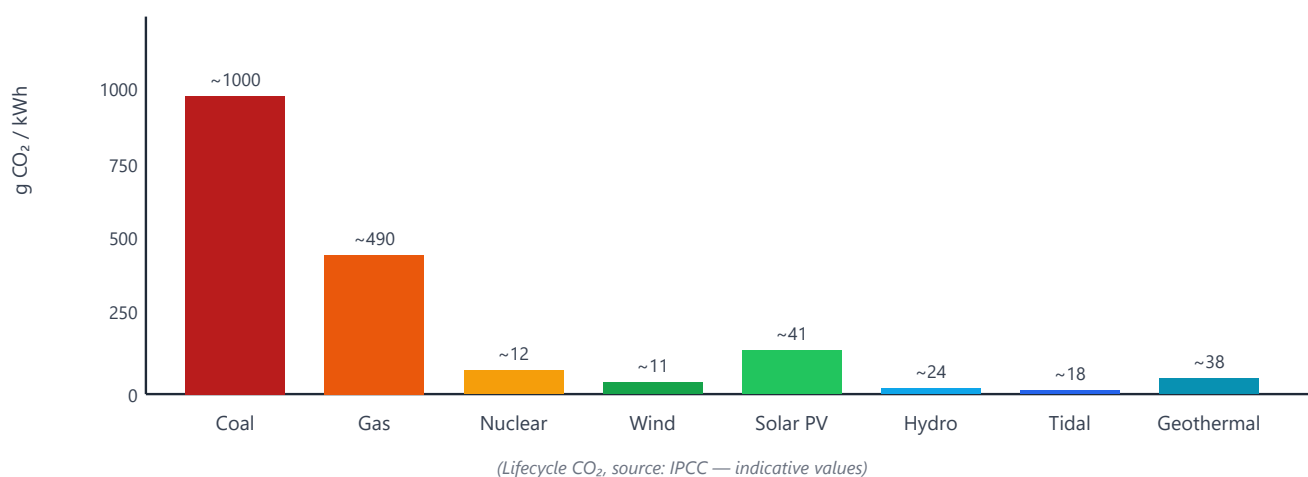
- **Non-renewable:** will run out because they are used faster than they form. Fossil fuels (coal, oil, gas) and nuclear fuel (uranium).
- **Renewable:** replenished naturally faster than they are used. Wind, solar, hydroelectric, tidal, wave, geothermal, biofuels.

Resource	How it's used	Advantages	Disadvantages
Coal / oil / gas	Burnt to boil water → steam turns turbine → generator	Reliable, high energy density, established infrastructure	CO ₂ → climate change; SO ₂ → acid rain; non-renewable
Nuclear (fission of uranium)	Heat from fission boils water → turbine → generator	No CO ₂ ; very large energy per kg of fuel; reliable	Radioactive waste (very long-lived); high decommissioning cost; accident risk
Wind	Wind turns turbine blades → generator	No fuel, no CO ₂ , free	Intermittent (no wind = no power); visual/noise impact on landscape
Solar (PV)	Photovoltaic cells convert sunlight directly to electricity	No fuel, no CO ₂ , modular	Only works in daylight; efficiency drops on cloudy days
Hydroelectric	Water stored behind a dam turns a turbine as it falls	Reliable, can ramp up quickly, no fuel	Drowns valleys (habitat loss), methane from rotting vegetation, very expensive

Tidal	Tidal movement turns turbines (e.g. estuary barrage)	Predictable, no fuel, no CO ₂ at point of use	Only certain coastal locations; disrupts estuary ecology
Wave	Rising/falling water drives generators	Renewable, no fuel	Hazard to shipping; variable output
Geothermal	Uses heat from hot rocks to boil water	Reliable, small footprint, low CO ₂	Few suitable locations (Iceland, NZ, parts of Italy)
Biofuel (ethanol, biodiesel)	Burn plant/animal matter or ferment to alcohol	Roughly carbon-neutral if crop is re-grown	Competes with food crops for land; deforestation in some cases

Comparing Resources

Typical CO₂ emissions per kWh electricity generated



P2.7 Trends in Electricity Generation

The UK electricity mix has shifted dramatically since 1990:

- Coal: fell from > 70% of generation to almost zero by 2024.
- Gas: rose through the 1990s; still the largest single source.

- Renewables (wind + solar + biomass + hydro): rose from ~2% to ~40%+.
- Nuclear: steady, around 15%.

Exam Tip: A standard 6-mark question: "Compare the environmental and economic advantages/disadvantages of two named energy resources." Always name **specific** drawbacks (CO₂, acid rain, habitat loss, radioactive waste) — not generic "bad for environment".

P2.8 Power Stations & the National Grid (overview)

Thermal Power Station Energy Flow



Waste heat is lost to the surroundings (often via cooling towers) — this limits efficiency to $\approx 35\text{--}45\%$.

Exam Tip: Wind, hydro, tidal and wave skip the "heat $\rightarrow$ steam" step — they drive turbines directly, giving higher overall efficiency.